Gravity from Quantum Measurement

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March 31, 2026

Abstract

We propose a framework in which gravity can emerge from the physical backreaction of quantum measurement and the microstate dynamics of information acquisition. Standard results, including horizon scales and entropy bounds, follow from Landauer's principle at the Unruh temperature. We then present a concrete bilocal microstate construction as a case-study. Entangled pairs emitted from a localized source propagate into left and right Rindler wedges, where polarization measurement events produce discrete energy-momentum impulses and correlated classical bits. These microstates can be described geometrically using Aichelburg–Sexl pp-wave solutions, whose longitudinal shifts and transverse focusing imprint energy and information directly onto horizon geometry. This setup illustrates how individual measurement events contribute to horizon entropy, and how entanglement can be converted into spatial information, providing an operational realization of ER=EPR at the level of single microstates. The framework complements entropic and thermodynamic approaches by supplying a concrete, microphysical model linking quantum measurement, horizon dynamics, and emergent spacetime.

1 Introduction

Two landmark results establish that gravity and thermodynamics are deeply intertwined. Jacobson showed that the Einstein equations can be derived from thermodynamic identities applied to local Rindler horizons, with horizon entropy playing the role of the fundamental quantity from which spacetime dynamics emerge [1]. Verlinde extended this perspective, arguing that gravitational forces arise from entropy gradients on holographic screens, recasting gravity as an entropic rather than a fundamental interaction [2]. Both frameworks, however, operate at the level of bulk thermodynamics and leave open a prior question: what physical process underlies each individual entropy increment?

We argue that the answer is quantum measurement. Each act of information acquisition by a physical observer constitutes a discrete microstate event, and the collective backreaction of such events—together with the causal correlations they induce—gives rise to gravitational effects. The argument proceeds in three steps. First, any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. Second, by the equivalence principle, this impulse is locally indistinguishable from the effect of a gravitational field: no local measurement can distinguish the recoil associated with a quantum measurement event from thrust in a gravitational potential. Third, when such impulses are modelled geometrically—via Aichelburg–Sexl (AS) pp-wave solutions, as we show—they imprint energy and information directly onto horizon geometry,

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and coarse-graining over the resulting ensemble recovers standard horizon thermodynamics.

A unifying structure runs through all three of these relations, and it is worth making explicit before proceeding. In each case, the relation between two adjacent concepts has the same logical character: every instance of the more abstract concept is grounded in an instance of the more physical one, but not every instance of the more physical concept gives rise to the more abstract one *for a given observer*. The gap in each case is filled by subjectivity. Quantum state reduction is always backed by an observation—it just is a Bayesian update on the observer's part—but an observation that carries no new information for that observer produces no state reduction. Equally, every observation is backed by an energy-momentum transfer, but not every such transfer constitutes an observation for you: an interaction involving a different observer leaves your state unchanged. The Unruh effect suggests the same structure extends to the equivalence principle: an accelerating observer perceives a thermal bath and attributes it to a gravitational field, while an inertial observer sees only vacuum. Whether an acceleration counts as gravity is perspectival in precisely the same sense. In each case the relation is an identification, not a causal link, and the identification is observer-relative.

This stands in sharp contrast to the Penrose–Diósi objective collapse programme [3, 4], which postulates that gravity *causes* quantum state reduction as an objective, observer-independent physical process. Figure 1 summarises this contrast. The three solid arrows represent the perspectival grounding relations described above; the single dashed arrow represents the Penrose–Diósi causal postulate. It is the only relation in the diagram claimed to be objective and observer-independent, and it is the only one we do not adopt.

$$\begin{tikzcd}
\text{Gravitational field} \arrow[r, dashed, "\text{Penrose–Diósi}"] & \text{Quantum state reduction} \\
\text{Acceleration} \arrow[u, "\text{equivalence principle}"] \arrow[r, "\text{energy–momentum transfer}"'] & \text{Observation} \arrow[u, "\text{Bayesian update}"']
\end{tikzcd}$$

Figure 1: Solid arrows denote substrate relations: $A \to B$ means every instance of B is physically realised by an instance of A, for a given observer. The dashed arrow denotes the Penrose–Diósi postulate, where the relation is claimed to hold objectively rather than observer-relatively.

To make this concrete, we build on the microstate construction of [5], in which bilocal driving sources generate entangled Rindler excitations that realise individual terms of the Unruh thermal ensemble. The present paper applies and extends that construction. Entangled pairs emitted from a localised source propagate into left and right Rindler wedges; polarisation measurement events produce discrete energy-momentum impulses and correlated classical bits. We show that each such microstate carries a calculable gravitational imprint, modelled by an AS pp-wave solution, and that coarse-graining over the ensemble recovers standard horizon thermodynamics. This construction also provides an operational realisation of ER=EPR at the level of individual microstates: entanglement is converted into geometric information, and each measurement event contributes a discrete, calculable increment to horizon entropy.

This approach both complements and sharpens the thermodynamic gravity programme of Jacobson and Verlinde. Jacobson's derivation of the Einstein equations from horizon thermodynamics and Verlinde's entropic force framework both operate at the level of

bulk averages, leaving open the question of what physical process underlies each entropy increment. The present construction supplies that microstate realisation: each entropy increment corresponds to a physically mediated measurement event with a calculable energy-momentum transfer. Thermality is emergent rather than intrinsic, arising from coarse-graining over an ensemble of source-driven microstates, each of which admits a definite, non-thermal description in the global Minkowski frame. The thermodynamic results of both programmes are recovered as a consequence of this coarse-graining, rather than assumed as a starting point.

[Section map paragraph todo]

2 Motivating Heuristics

The central observation is simple. Landauer's principle associates a minimal energy cost with the acquisition of information [6]. For an accelerated observer, the relevant temperature is the Unruh temperature [7]. Substituting one into the other eliminates temperature entirely, yielding a frame-independent relation between information acquisition, energy transfer, and acceleration (a version of the BHT first law). This is not merely an algebraic convenience: it is a concrete dictionary between the thermal description natural to a Rindler observer and the non-thermal global Minkowski description, with the apparent thermality of the vacuum emerging as a consequence of the observer's acceleration rather than as a fundamental property of the state. In this way we derive a sequence of standard results in black hole thermodynamics.

2.1 The Unruh-Landauer Relation

Landauer's principle states that the acquisition of ΔS bits of information by a physical observer requires a minimum energy transfer

$$\Delta E \geq k_B T \, \Delta S. \tag{1}$$

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta S.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a frame-independent relation between energy, acceleration, and information, with no reference to any thermal description. It can be read as the first law of black hole thermodynamics derived from below — from the physical cost of information acquisition — rather than postulated as a thermodynamic identity. In what follows, all standard results emerge as consequences of this single relation. We note that Landauer's principle is conventionally stated for information erasure rather than acquisition. Here we apply it to acquisition, which is the physically relevant process for an observer registering a measurement outcome. The bound has the same form in both cases; we apply it to acquisition on the grounds that registering a measurement outcome is physically equivalent to a bit being written into the detector's state, which is the process Landauer's principle governs.

2.2 Momentum Transfer and Acceleration

The energy transfer ΔE implied by equation (3) is carried by a physical quantum — matter or radiation — mediating the measurement interaction. We refer to this as the

thrust quantum. For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{4}$$

Let the interaction occur over a spatial scale r, so that the interaction time is $\Delta t \sim 2r/c$. For a detector of mass m_d, Newton's second law in impulse form gives

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2 m_d r}. \tag{5}$$

The factor of 2 in the interaction time is not an independent assumption: it will be fixed self-consistently by requiring that the characteristic interaction scale reproduces the Schwarzschild radius, as shown in the next subsection.

2.3 The Schwarzschild Radius

The thrust quantum carries energy ΔE and produces an acceleration a on the detector. Identifying this with the Newtonian gravitational acceleration due to a mass m_t at distance r.

$$a = \frac{G m_t}{r^2}, \tag{6}$$

and substituting equation (5) with $\Delta E = m_t c^2$, we obtain

$$\frac{m_t c^2}{2 m_d r} = \frac{G m_t}{r^2}. \tag{7}$$

Solving for r:

$$r = \frac{2 G m_d}{c^2} \equiv r_s, \tag{8}$$

which is precisely the Schwarzschild radius of the detector mass m_d. This fixes the factor of 2 in the interaction time retroactively: the choice $\Delta r = 2r/c$ is the unique one consistent with Newtonian gravity and the correct Schwarzschild scale.

2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{G m_d}{r_s^2} = \frac{c^4}{4 G m_d}, \tag{9}$$

in agreement with the standard Schwarzschild result. This is the acceleration that appears in the Unruh temperature associated with the detector's horizon, $T = \hbar\kappa / 2\pi k_B c$, providing an internal consistency check on the framework.

2.5 Bekenstein-Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity κ for a, the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta S \leq \frac{2\pi c}{\hbar \kappa} \Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot c^2 \Delta m_d. \tag{10}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta S \leq \frac{\Delta A}{4 \ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{11}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

The two bounds are saturated simultaneously when $a = c^2/R$ — which is exactly the condition for being *at a horizon*. The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the minimum cost of information acquisition equals the maximum information content of the region.

2.6 Summary

Beginning from the single relation (3), and using only Newton's law and the equivalence principle, we have recovered the Schwarzschild radius, the surface gravity, and the Bekenstein-Hawking area law as consequences of the observation=acceleration identification. These are standard results, but their derivation here from the physical cost of information acquisition rather than from thermodynamic postulates is the point. The thrust quantum introduced in this section — the physical carrier of the measurement-induced impulse — will be given a precise microstate realization in the sections that follow, where it is identified with an individual photon modeled as an Aichelburg-Sexl pp-wave.

3 Bilocal Microstate Construction

The heuristic arguments of Section 2 establish a relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. Following [5], we invert this logic: we place an interaction in the past, where a source injects non-thermal entangled excitations into the field, and show that the observer's accelerated response is driven by these source-induced excitations. The thermal ensemble then emerges as a coarse-grained aggregate of such microstates, rather than being a primitive property of the vacuum.

We now upgrade this construction from a spin-0 scalar field to the spin-1 electromagnetic field. This is not merely a technical generalisation. As we show, the photon carries a unit of action $\hbar$ per cycle alongside its energy $E = \hbar\omega$, which closes the Landauer gap left open in Section 2: the minimum actionable event in the construction is precisely a single photon, and its polarisation state constitutes the classical bit acquired by each observer. The anticorrelated helicities of the entangled pair are the natural completion of the $\mathbb{Z}_2$ swap symmetry between wedges, and as we discuss in Section 4, they imprint a definite geometric signature onto the horizon consistent with the ER=EPR identification.

3.1 The Paz–Lester–Wright Setup

We retain the Alice-and-Bob style naming convention. Three observers are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named *Wright* (R), *Lester* (L), and *Paz* (P) respectively — a mnemonic for Right, Left, and Past. Paz serves as the common causal ancestor of Wright and Lester (see Figure 2).

Paz emits an entangled pair of spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The momenta are perfectly balanced, so the source exhibits no net back-reaction in this minimal model — a simplification discussed further in Section **??**. The two photons propagate into the left and right Rindler wedges, where Lester and Wright detect them respectively, each registering a helicity eigenvalue and acquiring one classical bit of information.

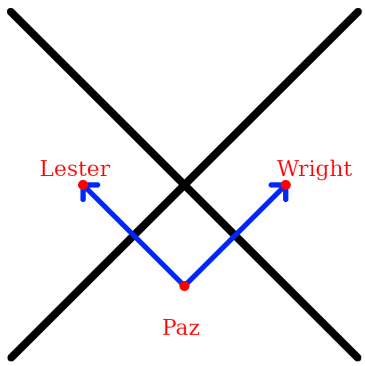

Figure 2: A source in the past, Paz (P), emits an entangled photon pair that accelerates Wright (R) into the right Rindler wedge and Lester (L) into the left Rindler wedge. The two null directions of propagation define the x^+ and x^- axes of the double-null coordinate system.

3.2 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector k_R^μ and the left-moving photon has wave-vector $k_L^\mu = -k_R^\mu$. These two null directions are precisely the x^+ and x^- axes of double-null coordinates

$$x^\pm = \frac{1}{\sqrt{2}}(t \pm z), \tag{12}$$

with transverse coordinates $\mathbf{x}_\perp = (x^1, x^2)$. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(x) = 0, \qquad A_-(x) = 0, \tag{13}$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^1, A^2)$. This gauge respects the $\mathbb{Z}_2$ symmetry that swaps the two wedges ($x^+ \leftrightarrow x^-$) simultaneously with ($R \leftrightarrow L$), and therefore treats both observers symmetrically by construction. The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon^i_{(\lambda)}$ with $\lambda = \pm 1$.

The lack of manifest four-dimensional covariance in (13) is a consequence of aligning the gauge with the pair of physical null directions selected by the source, rather than an arbitrary choice. The full Lorentz group is broken to the little group preserving the pair $\{k_R^\mu, k_L^\mu\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane. This is precisely the little group of massless particles, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice.

3.3 The Bilocal Source

In double-null gauge the electromagnetic field is characterised by its transverse components alone. The bilocal driving source generalises the scalar construction to

$$J^{ij}(x, y) = \lambda\, \epsilon^{i*}_{(-1)}\, \epsilon^j_{(+1)}\, f_L(x)\, f_R(y), \tag{14}$$

where λ is a coupling constant, f_L and f_R are normalised mode profiles supported on the left and right wedges respectively, and $\epsilon^i_{(\lambda)}$ are the circular polarisation vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}. \tag{15}$$

The index structure of (14) assigns helicity $+1$ to the right-moving mode and helicity -1 to the left-moving mode: the pair is *anticorrelated* in helicity. This is the unique assignment consistent with angular momentum conservation at the emission vertex and with the $\mathbb{Z}_2$ wedge-swap symmetry, under which $\epsilon^i_{(+1)} \leftrightarrow \epsilon^i_{(-1)}$.

The sourced Lagrangian reads

$$\begin{aligned}\mathcal{L}_{\text{sourced}} &= \mathcal{L}_{\text{free}} + \tfrac{1}{2}\lambda \int J^{ij}(x,y)\, A^+_i(x)\, A^+_j(y)\, d^4x\, d^4y \;+\; \text{h.c.} \\ &= \mathcal{L}_{\text{free}} + \tfrac{1}{2}\lambda \Big(a^{L\dagger}_{-k,(-1)}\, a^{R\dagger}_{k,(+1)} \;+\; a^{L}_{-k,(-1)}\, a^{R}_{k,(+1)} \Big),\end{aligned} \tag{16}$$

where $a^{L\dagger}_{-k,(-1)}$ and $a^{R\dagger}_{k,(+1)}$ are creation operators for left-wedge helicity -1 and right-wedge helicity $+1$ modes respectively. The interaction term in (16) is a two-mode squeezing interaction at frequency ω, now resolved by helicity: it is the direct vector-field analogue of the scalar squeezing interaction [**?**], with the anticorrelated polarisation structure made explicit. The photon pair carries total angular momentum zero, consistent with emission from a scalar source at Paz.

Each photon carries energy $E = \hbar\omega$ and, as a mode of the electromagnetic field, constitutes precisely one quantum of action $\hbar$ per oscillation cycle. The thrust quantum of Section 2 is thereby identified with a single photon: the minimum actionable event is the emission and detection of one such quantum. The energy transferred per bit acquired is $\Delta E = \hbar\omega$, saturating the Unruh–Landauer bound (3) at the fundamental quantum level. This closes the Landauer gap: registering a photon's helicity writes one classical bit into the detector's physical state, which is precisely the process Landauer's principle governs.

3.4 Polarisation Bogoliubov Transformation

We now show that the polarisation labels survive the Rindler–Unruh mode decomposition, so that the thermal ensemble retains its helicity structure and the anticorrelation of the emitted pair descends unambiguously to the level of individual Rindler microstates.

In the transverse plane, the electromagnetic field in double-null gauge decomposes into two independent scalar-like sectors labelled by helicity $\lambda = \pm 1$. For each sector, the right-wedge field expands as

$$A^i_\lambda(x) = \int_0^\infty \frac{d\Omega}{2\pi} \frac{1}{\sqrt{2\Omega}}\, \epsilon^i_{(\lambda)} \left[b^R_{\Omega,\lambda}\, u^R_\Omega(x) + b^{R\dagger}_{\Omega,\lambda}\, u^{R*}_\Omega(x) \right], \tag{17}$$

where $u^R_\Omega(x)$ are the right-wedge Rindler modes at Rindler frequency Ω, and $b^R_{\Omega,\lambda}$, $b^{R\dagger}_{\Omega,\lambda}$ are the corresponding annihilation and creation operators satisfying

$$\left[b^R_{\Omega,\lambda},\, b^{R\dagger}_{\Omega',\lambda'} \right] = \delta(\Omega - \Omega')\, \delta_{\lambda\lambda'}. \tag{18}$$

An identical expansion holds in the left wedge with $R \to L$ and $\lambda \to -\lambda$, reflecting the anticorrelated helicity assignment. Because each helicity sector is independently a scalar-like sector, the Bogoliubov transformation relating Minkowski and Rindler modes takes the same form in each sector. For helicity λ at Rindler frequency Ω, the Unruh modes are

$$c_{\Omega,\lambda} = \cosh\theta_\Omega\, b^R_{\Omega,\lambda} - \sinh\theta_\Omega\, b^{L\dagger}_{\Omega,-\lambda}, \tag{19}$$

where $\tanh\theta_\Omega = e^{-\pi\Omega c/a}$ encodes the Unruh distribution at acceleration a, and $-\lambda$ on the left-wedge operator reflects the anticorrelation. The transformation (19) is the standard Bogoliubov relation for a massless field [7, **?**], now carrying a helicity label throughout. The Minkowski vacuum in the Rindler basis reads, for each helicity sector,

$$|0\rangle^{(\lambda)}_M = \frac{1}{\cosh\theta_\Omega} \sum_{n=0}^{\infty} \tanh^n\theta_\Omega\, |n_{\Omega,\lambda}\rangle_R\, |n_{\Omega,-\lambda}\rangle_L, \tag{20}$$

and the full vacuum is the product over all Ω and both helicities:

$$|0\rangle_M = \prod_{\Omega,\lambda} |0\rangle_M^{(\lambda)}. \tag{21}$$

Equation (20) makes explicit that the Minkowski vacuum is a sum over entangled helicity-anticorrelated pairs at each Rindler frequency. The bilocal source selects a single term from this sum at fixed Ω and fixed helicity pair $(\lambda, -\lambda) = (+1, -1)$, yielding the single-microstate

$$|\Psi_\Omega\rangle_M = |1_{\Omega,+1}\rangle_R \, |1_{\Omega,-1}\rangle_L + \mathcal{O}(\text{higher order}). \tag{22}$$

Wright's detection of helicity $+1$ and Lester's detection of helicity -1 are therefore perfectly anticorrelated: each measurement event yields one classical bit, and the two bits are the complement of one another. Table **??** summarises the correspondence between mode profiles, operators, helicity labels, and states.

The key structural result is that the helicity label λ is conserved under the Bogoliubov transformation: it enters equation (19) as a spectator index and exits unchanged on the Unruh operators $c_{\Omega,\lambda}$. Thermality, when it arises from coarse-graining over the ensemble of microstates (22), is therefore helicity-resolved: each helicity sector thermalises independently, and the anticorrelation between wedges is preserved at every level of the construction.

3.5 Summary of the Vector Upgrade

The passage from the scalar to the vector field construction introduces three structural improvements. First, the photon carrying $E = \hbar\omega$ and one quantum of action per cycle identifies the thrust quantum of Section 2 with a concrete physical object, closing the Landauer acquisition gap. Second, the double-null gauge condition, motivated by the pair of null directions defined by the emission event, treats both Rindler wedges symmetrically and exposes the $\mathbb{Z}_2$ swap symmetry as the organising principle of the construction. Third, the anticorrelated helicity structure, conserved through the Bogoliubov transformation, provides each observer with one classical bit whose values are perfectly complementary: the two bits together constitute the correlated classical record that, in the AS pp-wave picture of Section 4, is imprinted as geometric information onto the horizon.

3.6 Matter as Substrate: Ticks, Memory, and the Thrust Quantum

The claim that information is encoded in matter and energy rather than in the horizon geometry requires a physical account of what it means for matter to *carry* information. We outline that account here, drawing on the matter channel picture of [**?**].

A massive particle accumulates phase along its worldline at a rate set by its rest energy. The De Broglie relation $f = E/h$ assigns a natural frequency to any mass m, and Bremermann's computational limit independently recovers the same rate $f_{Br} = mc^2/h$ as the maximum clock rate of any physical system with that energy [**?**]. These two results agree because they are both expressions of the same underlying fact: a quantum of action $\hbar$ is the minimal unit of physical evolution, and mc^2/h is the rate at which a mass m consumes such quanta. Each such quantum corresponds to one tick of the phase $e^{iS/\hbar}$ in the path integral, and classical trajectories emerge from the constructive interference of these ticks across paths.

This gives matter a precise sense in which it is a *substrate for information*: it accumulates ticks, and each tick is a discrete, Lorentz-invariant event recorded in the particle's proper time. Matter is persistent memory in the operational sense that its history of ticks is physically encoded in its state and cannot be erased without an energy cost governed

by Landauer's principle. The maximum rate of this memory is bounded by Bremermann; the maximum total capacity is bounded by the Bekenstein bound, which we have already derived in Section 2 as a consequence of the Unruh–Landauer relation rather than as an independent postulate.

A massless particle — the photon — has no rest frame, no proper time, and accumulates no action between emission and absorption. It does not tick. It therefore carries no persistent memory and cannot serve as a substrate. Its role in the present construction is precisely that of a pure information carrier. A single photon carries exactly two irreducible constituents, distinct in kind:

- **One quantum of action** $\hbar$: the physical transfer, deposited into the detector's matter-memory as one additional tick. This is what makes the interaction irreversible and imposes the Landauer energy cost.
- **One bit of polarisation**: the informational outcome, the classical record of which helicity — $\lambda = +1$ or $\lambda = -1$ — was absorbed. This is what is correlated between Wright and Lester and constitutes the spatial bit of Section 3.7.

These are not the same thing in different guises. The $\hbar$ is what makes the event physical; the bit is what makes it meaningful to an observer. That a single photon carries exactly one of each is not a coincidence: it is why the photon is the natural minimal unit of the entire construction, the unique object that saturates both the action and the information count simultaneously. The energy deposited is $\Delta E = \hbar\omega$, saturating the Unruh–Landauer bound at the fundamental quantum level, as established in Section 3.

The horizon enters this picture not as a storage medium but as a boundary condition. It marks the surface at which the matter-memory on one side is causally disconnected from the matter-memory on the other. Horizon entropy, in this account, is not a property of the horizon surface itself but a count of the ticks that have been written into the matter on each side — bounded by Bremermann from above and by Landauer from below. The holographic appearance of this count, proportional to area rather than volume, is a consequence of the Unruh–Landauer and Bekenstein relations derived in Section 2 and does not require holography as an independent postulate.

This reframing has a direct consequence for the ER=EPR identification developed in the next subsection. The correlated measurement outcomes of Wright and Lester are not encoded in the horizon geometry in the first instance — they are encoded in the matter of their respective detectors, each of which has absorbed one photon and accumulated one additional tick. The geometric correlations between the two wedges, realised as correlated AS pp-wave imprints in Section 4, are downstream of this matter-level encoding. The bridge, if it exists, is built from correlated ticks, not from the horizon itself.

3.7 Entanglement as Spatial Information: ER=EPR at the Microstate Level

Each measurement produces a polarisation outcome — H or V — at each wedge, perfectly anticorrelated between Lester and Wright. But consider a different question: not *what* was observed, but *where*. The question "who observed H?" has a binary answer: Lester (left wedge) or Wright (right wedge). This is a *spatial bit* — a bit that encodes not a polarisation state but a location in space.

Each measurement event has, as established in Section 3.6, exactly two irreducible constituents: one quantum of action $\hbar$ written irreversibly into the detector's matter-memory as a tick, and one bit of polarisation registered as a classical outcome. The entanglement between Wright and Lester is consumed in producing these two records — one tick and one bit at each wedge. What remains after measurement is not a remnant of the entanglement but its classical residue: the perfect anticorrelation between the two bits, and the correlated pair of ticks written into the matter on each side of the horizon.

The spatial bit is precisely the polarisation bit of Section 3.6 now read as a location rather than a spin state: the same bit that records *what* was observed also records *where*, because in this construction the two wedges carry opposite helicities by design.

The information is carried by the matter of the detectors, not by the horizon. As established in Section 3.6, the horizon is a boundary that makes the left/right distinction sharp; it is the matter on each side that accumulates the ticks and holds the classical record. Wright's detector has absorbed one photon, gained one tick, and registered helicity $+1$. Lester's detector has absorbed one photon, gained one tick, and registered helicity -1. The two records are perfectly complementary by the anticorrelated structure of the bilocal source, and their complementarity is encoded in matter before it is reflected in geometry.

This reframing sharpens the ER=EPR identification. The two wedges are connected not by the horizon as an abstract geometric object but by the physical correlations between the energy-momentum impulses deposited in the matter on each side. In the AS pp-wave picture developed in Section 4, each such impulse produces a measurable longitudinal shift and transverse focusing of the horizon geometry. The correlated pair of impulses — one in each wedge, anticorrelated in helicity — produces correlated geometric deformations on the two sides of the horizon, and it is *this* geometric correlation, downstream of the matter-level encoding, that constitutes the Einstein–Rosen bridge at the level of a single microstate.

The spatial bit — the answer to "who observed H?" — is therefore not merely a record of which side the photon landed on. It is a label for which of the two correlated geometric deformations was realised, and the two labels together reconstruct the bridge. This is an operational realisation of ER=EPR at the level of individual microstates: the entanglement between Lester and Wright's outcomes is not a stand-in for the bridge but is physically instantiated in the pair of correlated ticks whose geometric imprint *is* the bridge. The bridge is built from matter, tick by tick.

4 Gravitational Imprint: the pp-Wave Realisation

Each measurement event in the bilocal construction involves a photon transferring energy and momentum to an accelerated detector. We now model the gravitational imprint of this transfer by identifying each photon with an Aichelburg–Sexl (AS) pp-wave solution in general relativity [**?**]. The AS geometry is an exact solution describing the gravitational field of an ultrarelativistic point mass, and provides the natural geometric language for a single-photon microstate.

4.1 The AS pp-Wave and Its Effects

Let $u = t - z$ and $v = t + z$, where the photon propagates along the null plane $u = 0$, z is the longitudinal direction, and (x, y) are transverse coordinates. Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by
$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{23}$$
This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.
2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity
$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{24}$$

directed toward the axis. Test particles are pulled toward the centre of the wave, regardless of their initial transverse separation.

4.2 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by Paz propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $-v$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction.

This is not merely a kinematic effect. The transverse plane of the pp-wave is identified with horizon area elements $dA = dx\,dy$, and the longitudinal shift creates a new null interval between the two wedges — a region of spacetime that did not exist, in the sense of being causally accessible, before the photon passed. The bit of spatial information encoded in the left/right measurement outcome — "who observed H?" — refers to precisely this separation. The measurement event does not merely *record* a spatial distinction; through the gravitational backreaction of the photon, it *creates* the spatial separation that the bit describes.

Put differently: the same physical event — a photon crossing a horizon — simultaneously encodes a classical bit and, via the AS longitudinal shift, generates the spatial structure that bit refers to. Space is not a pre-existing backdrop onto which information is written; it is produced, increment by increment, by the gravitational imprint of each measurement event. This is the microstate-level content of emergent space: each detection event contributes both an entropy increment and a spatial increment, and the two are the same thing described in different languages.

4.3 Horizon Entropy from Individual Microstates

Each detection event contributes a discrete entropy increment via the Bekenstein–Hawking relation derived in Section 2:

$$\Delta S = \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{25}$$

where ΔA is the area element imprinted on the horizon by the transverse focusing of the pp-wave. Horizon entropy is therefore not a bulk thermodynamic quantity assigned to the horizon as a whole, but a sum of discrete, calculable contributions from individual measurement-induced interactions. The transverse focusing and the longitudinal shift are two aspects of the same microstate: focusing encodes the entropy increment in the area; the longitudinal shift encodes the spatial increment in the geometry.

4.4 Coarse-Graining and the Emergence of Thermality

In a steady-state scenario, Paz continuously emits entangled pairs, and Lester and Wright continuously detect them. Each detection is an AS pp-wave microstate with a definite, non-thermal description in the global Minkowski frame: a specific photon energy, a specific transverse position, a specific longitudinal shift. Coarse-graining over this ensemble recovers the Unruh thermal spectrum at temperature $T = a/2\pi$, consistent with the vacuum decomposition of equation (**??**). The thermodynamic description of the horizon emerges from this averaging; it is not a feature of the individual microstates.

This completes the microstate programme. Each entropy increment in the Bekenstein–Hawking area law corresponds to a single AS pp-wave interaction. Each pp-wave interaction corresponds to a single measurement event in the bilocal construction. Each measure-

ment event both encodes a spatial bit and generates the spatial separation that bit refers to. The gravitational, thermodynamic, information-theoretic, and geometric descriptions of the horizon are unified at the level of individual microstates, and thermality, entropy, and space itself emerge together from the coarse-grained aggregate of these events.

5 Conclusion

6 Acknowledgements

I'm also grateful to Claude and ChatGPT for helpful discussions.

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